

CS 577 Deep Learning

Homework 1: Tools and Tensor Fluency

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Part 1: LaTeX and Mathematical Communication

1.1 Tensor notation

Typest the following statement cleanly, including dimensions:

For a batch X with shape (B, D) , weights W with shape (D, K) , and bias b with shape $(K,)$, the affine output is

$$Y = XW + 1_B b^T$$

and has shape (B, K)

Here, $1_B \in \mathbb{R}^{B \times 1}$ denotes the all-ones column vector. Therefore, $1_B b^T \in \mathbb{R}^{B \times K}$ repeats the bias vector b^T once for each of the B examples. This is the mathematical form of broadcasting b across the batch dimension in the NumPy/PyTorch expression

$$Y = XW + b$$

Use mathematical notation rather than copying the monospace statement verbatim.

1.2 Regularized mean-squared error

Typest the objective

$$J(w) = \frac{1}{N} \|Xw - y\|_2^2 + \frac{\lambda}{2} \|w\|_2^2$$

and its gradient with respect to w

1.3 Results table and interpretation

Quantity	Value
Initial training loss	TODO
Final training loss	TODO
Learned slope	TODO
Learned intercept	TODO
Maximum gradient difference	TODO

Table 1: Results from the deterministic regression and gradient checks.

Part 2: NumPy Tensors and Vectorization

2.1 Shapes and broadcasting

shapes: $X.shape=(4,3)$ $W.shape=(3,2)$ $b.shape=(2,)$ $Y.shape=(4,2)$

$X@W$ is matrix multiplication. Since X has shape $(4,3)$ and W has shape $(3,2)$, the inner dimensions match ($3=3$), so the result has shape $(4,2)$. The bias b has shape $(2,)$. Broadcasting compares dimensions from right to left, so $(2,)$ is treated like $(1,2)$. Since $2=2$ and 1 can expand to 4, b is broadcast to shape $(4,2)$ and added to every row of $X@W$.

$$Y = \begin{bmatrix} 4.25 & -3.75 \\ 0.25 & 5.25 \\ -2.25 & -0.75 \\ -1.25 & 4.25 \end{bmatrix}$$

2.2 Standardization

mean:

$$[0.5, 0.25, 1.25]$$

std:

$$[1.11803399, 1.47901995, 1.47901995]$$

check mean:

$$[0.0, -1.38777878 \times 10^{-17}, 0.0]$$

check std:

$$[1, 1, 1]$$

2.3 Pairwise squared distances

$$D2 = \begin{bmatrix} 2 & 1 \\ 1 & 4 \\ 2 & 5 \end{bmatrix}$$

The inserted singleton dimensions allow broadcasting to compare every row of A with every row of B simultaneously, without a Python loop.

2.4 Vectorization timing

Part 3: PyTorch Tensors and a Minimal Model

3.1–3.2 Agreement, dtype, and device

Tensors participating in an operation generally need to be on the same device because CPU and GPU memory are separate. Their dtypes also need to be compatible; otherwise PyTorch may raise an error or perform a type conversion. Explicitly choosing both the device and dtype makes the computation predictable

3.3–3.4 Model and training

Part 4: Three Views of the Same Gradient

4.1 Analytic gradient

4.2–4.3 Numerical comparison

Part 5: Reflection

Shape/broadcasting mistake. TODO

Gradient validation method. TODO

Workflow change before HW2. TODO

Assistance and Generative-AI Disclosure

TODO